

MATH 127 Calculus for the Sciences

Lecture 29



Today's lecture

Last time

Curve sketching

This time

Course note coverage Section 4.1.1

Absolute maxima/minima

a.k.a Global maxima and minima

Absolute maxima and minima

Didn't we do this already? I talked about global max and min when I talked about local max and min. What else do you want me to say?

Definition

- An input $x = a$ is a **global maximum** of f if

$$f(a) \geq f(x) \text{ for all } x \text{ in the domain of } f(x).$$

- An input $x = a$ is a **global minimum** of f if

$$f(a) \leq f(x) \text{ for all } x \text{ in the domain of } f(x).$$

Well, turns out there's some theory you can learn in terms of locating the maxima/minima.

Absolute maxima and minima

Fermat's theorem says that a **local max/min** must be a .

This means to find the **local max/min**, we only need to look for them among .

I say that a **global max/min** must be a .

So to find global max/min, we only need to look for them among , which lie among the .

Absolute maxima and minima

Proposed method

We can list out all critical points, then the biggest one is the global max and the smallest one is the global min.

But this method does not work in general, what are the flaws?

Absolute maxima and minima

Correct method

Suppose we know $f(x)$ is **continuous** on a **closed interval** $[a, b]$.

We can list out critical points and the edge points, then the biggest one is the global max and the smallest one is the global min **on the interval** $[a, b]$.

Example If I have a function continuous on the interval $[0, 1]$, would the flaws we saw previously still occur?

Extreme value theorem

Theorem If f is *continuous* on a *closed interval* $[a, b]$, then f must have a global maximum with respect to $[a, b]$

Remark So this says f must have a global max, but what about a global min?

Closed interval method

Suppose $f(x)$ is a continuous function defined on a closed interval $[a, b]$. Then you can do the following to find its global max and min:

1. Find all critical points.
2. List out the output at the two edge points of $[a, b]$ and at those critical points.
3. The biggest ones are global max, and the smallest ones are global min.

Warning:

- This only works if f is **contunious** and defined on a **closed interval**
- There might be multiple points where f achieves global max/min.

Closed interval method

Example Find the global max and global min, including at which inputs they are achieved, of the function $f(x) = \sin\left(\frac{1}{x}\right)$ on the closed interval $\left[\frac{1}{100\pi}, \frac{2}{\pi}\right]$.